

DATA-DRIVEN RETAIL SALES OPTIMIZATION

Adventure Works

Week 4 - Case Study 1

Data Analytics Project Report

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GitHub: https://github.com/amankumar1ext-web/Case_Study/tree/main/W4_Case_Study_1

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1 Data Loading & Initial Validation

1.1 Data Loading

```
=====
DimCustomer
=====
<class 'pandas.DataFrame'>
RangeIndex: 18484 entries, 0 to 18483
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerKey            18484 non-null  int64
1   GeographyKey           18484 non-null  int64
2   CustomerName           18484 non-null  str
3   BirthDate              18484 non-null  datetime64[us]
4   MaritalStatus          18484 non-null  str
5   Gender                 18484 non-null  str
6   EmailAddress           18484 non-null  str
7   YearlyIncome           18484 non-null  int64
8   Education              18484 non-null  str
```

1-1:Source Dataset Structure Identified During Python Data Loading

- Shows the row and column counts of the eight source datasets loaded into Pandas.
- Confirms that the expected dimension and fact tables were successfully extracted for further processing.

1.2 Data Validation

key_validation						
	Dataset	Key	Total_Rows	Unique_Keys	Null_Keys	Duplicate_Keys
0	DimCustomer	CustomerKey	18484	18484	0	0
1	DimEmployee	EmployeeKey	21	21	0	0
2	DimGeography	GeographyKey	655	655	0	0
3	DimProduct	ProductKey	397	397	0	0
4	DimReseller	ResellerKey	701	701	0	0
5	DimSalesTerritory	SalesTerritoryKey	11	11	0	0

1-2:Dimension Primary-Key Validation

- Confirms that the candidate keys of all six-dimension tables are unique and non-null.
- Provides the initial basis for establishing dimension-table integrity before relationship validation.

1.3 Initial Data Quality Assessment

Initial validation was performed on all eight source tables using Python and Pandas before any cleaning or transformation was applied.

Key observations:

- All eight datasets were successfully loaded and their structures were inspected.
- Candidate primary keys in all six-dimension tables were unique and contained no null values.
- No complete duplicate rows were identified in the dimension tables or `FactInternetSales`.
- `FactResellerSales` contained two duplicate occurrences involving three completely blank records.
- Forty-four reseller-sales records had no `SalesOrderNumber`; 41 of these contained financial values but lacked transaction and dimensional identifiers.
- After excluding records without a valid order number, no duplicate `SalesOrderNumber` and `SalesOrderLineNumber` combinations were identified.
- Missing values were also identified in selected dimension attributes and other source fields and were retained for further investigation during the cleaning stage.
- No data was modified during this validation stage. The identified issues were documented for investigation and treatment during data cleaning.

Table 1-1:Source Dataset Structure

DATASET	ROWS	COLUMNS
DIMCUSTOMER	18,484	13
DIMEMPLOYEE	21	9
DIMGEOGRAPHY	655	6
DIMPRODUCT	397	7
DIMRESELLER	701	4
DIMSALESTERRITORY	11	4
FACTINTERNETSALES	60,398	13
FACTRESELLERSALES	60,899	14

1.4 Data Summary

Table 1-2:Initial Data Validation Summary

VALIDATION CHECK	RESULT
DIMENSION PRIMARY KEYS	Unique and non-null
EXACT DUPLICATE ROWS IN DIMENSIONS	None
EXACT DUPLICATE ROWS IN FACTINTERNETSALES	None
EXACT DUPLICATE OCCURRENCES IN FACTRESELLERSALES	2
ROWS PARTICIPATING IN EXACT DUPLICATE GROUPS	3
RESELLER RECORDS WITHOUT SALESORDERNUMBER	44
INCOMPLETE RESELLER RECORDS WITH FINANCIAL VALUES	41
DUPLICATE VALID TRANSACTION KEYS	0
RAW DATE FIELDS	Stored as strings

1.5 Data Investigation

```
Rows without SalesOrderNumber: 44
```

```
Rows with SalesOrderNumber missing AND SalesAmount present: 41
```

```
Rows with SalesOrderNumber missing AND TotalProductCost present: 41
```

```
Rows with SalesOrderNumber missing AND both financial fields present: 41
```

1-3:Investigation of Incomplete Reseller Sales Records

- Shows that 44 reseller-sales records lack a transaction number.
- 41 of these records contain financial values but do not contain the transaction and dimensional identifiers required for normal fact-table analysis.

2 Data Cleaning & Transformation

2.1 Data Cleaning

2.1.1 Treatment of Incomplete Reseller Sales Records

During data validation, 44 records in `FactResellerSales` were identified without a `SalesOrderNumber`. These records also lacked the product, reseller, employee, territory and date identifiers required for transaction-level analysis.

Although 41 of the 44 records contained `SalesAmount` and `TotalProductCost`, they could not be reliably associated with individual transactions or business dimensions. The records were therefore excluded from the cleaned transaction-level dataset pending source clarification.

Before cleaning: 60,899 rows

Records excluded: 44

After cleaning: 60,855 rows

FACT RESELLER SALES - CLEANING VALIDATION

Raw rows: 60,899

Rows without SalesOrderNumber: 44

Excluded rows: 44

Clean rows: 60,855

Missing SalesOrderNumber after cleaning: 0

Duplicate transaction keys after cleaning: 0

Exact duplicate occurrences after cleaning: 0

2-1: Validation of FactResellerSales Cleaning

- Shows the removal of 44 incomplete reseller-sales records from the working dataset.
- Confirms that the cleaned transaction-level dataset contains no missing transaction identifiers or duplicate transaction keys.

2-1:FactResellerSales Cleaning Validation

VALIDATION	RESULT
RAW RECORDS	60,899
RECORDS WITHOUT SALESORDERNUMBER	44
RECORDS EXCLUDED	44
RECORDS RETAINED	60,855
MISSING SALESORDERNUMBER AFTER CLEANING	0
DUPLICATE TRANSACTION KEYS AFTER CLEANING	0
EXACT DUPLICATE OCCURRENCES AFTER CLEANING	0

2.1.2 Removing Duplicates

The raw FactResellerSales dataset contained two exact duplicate occurrences involving three rows. Post-cleaning validation showed zero exact duplicate occurrences and zero rows participating in duplicate groups.

The duplicate records were contained within the incomplete records already excluded from the transaction-level dataset. Therefore, no separate duplicate-removal operation was applied.

```
RAW DATA
-----
Exact duplicate occurrences: 2
Rows involved in duplicate groups: 3

CLEAN DATA
-----
Exact duplicate occurrences: 0
Rows involved in duplicate groups: 0
```

2-2:Duplicate counts before vs after

- The duplicate records were part of the already-excluded incomplete records and therefore no additional duplicate-removal transformation was required.

2.2 Handling missing values

DIMPRODUCT - COLOR CLEANING VALIDATION

```
Missing Color before cleaning: 56
Missing Color after cleaning: 0
Products assigned 'Unknown': 56
Product rows: 397
Unique ProductKeys: 397
Duplicate ProductKeys: 0
```

2-3:DimProduct Missing Color Treatment and Validation

- Shows the replacement of 56 missing product-color values with the controlled category "Unknown".
- Confirms that all 397 product records and their unique product keys were retained.

2.2.1 Treatment of Missing Sales Territory Attributes

One record in DimSalesTerritory contained a valid SalesTerritoryKey and SalesTerritoryRegion, but its SalesTerritoryCountry and SalesTerritoryGroup values were missing.

Further validation showed that SalesTerritoryKey = 11 was not referenced by either FactInternetSales or the cleaned FactResellerSales dataset. Since the available source data did not provide sufficient evidence to determine the missing descriptive values, they were not inferred or replaced with fabricated values.

The territory record was retained to preserve the source dimension structure.

2.2.2 Validation of Employee Hierarchy

```
Employees before validation: 21
Unique EmployeeKeys: 21
Duplicate EmployeeKeys: 0
Missing ParentEmployeeKey: 1
```

2-4:Employee Hierarchy Validation

- Confirms that one missing ParentEmployeeKey belongs to the CEO-level employee.
- Shows that all non-null parent employee keys correspond to valid employees.

- Demonstrates that the employee dimension retains its hierarchical structure without introducing an artificial parent.

2.2.3 Validation of Carrier Tracking Number

The CarrierTrackingNumber field in FactInternetSales was completely unpopulated. All 60,398 records contained missing values and no non-null tracking numbers were available for recovery.

The column was retained in the dataset, but no values were artificially assigned. The field was treated as a source-data limitation and excluded from analytical calculations.

Total Internet Sales records: 60,398

Missing tracking numbers: 60,398

Non-missing tracking numbers: 0

FACTINTERNETSALES - CARRIER TRACKING VALIDATION

Total rows: 60398

Missing CarrierTrackingNumber: 60398

Non-missing CarrierTrackingNumber: 0

Unique non-null tracking numbers: 0

2-5:FactInternetSales Carrier Tracking Number Validation

- Confirms that all 60,398 Internet Sales records have missing tracking numbers.
- Shows that no non-null tracking number was available for recovery.

3 Exploratory Data Analysis

3.1 EDA Objective

Exploratory Data Analysis was performed on the cleaned datasets to understand sales performance, cost structure, profitability, distributions and major patterns across the available sales channels.

The analysis was initially performed separately for Internet and Reseller sales to preserve the characteristics of each channel.

3.2 Channel-Level Overview

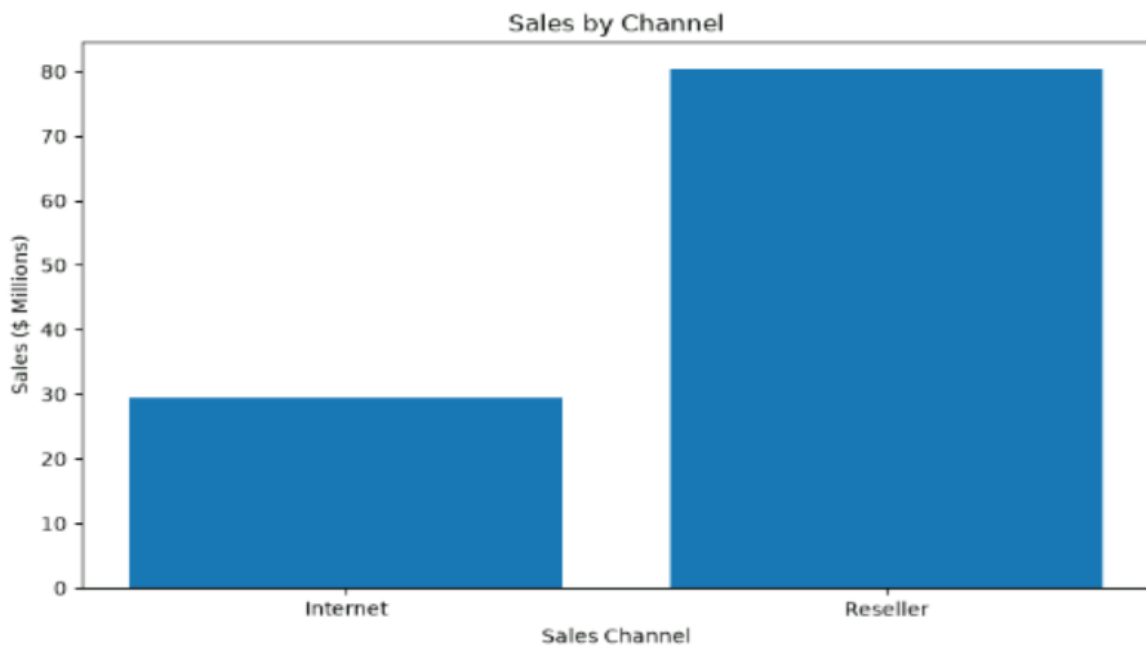
The initial analysis calculated sales revenue, product cost, gross profit and gross margin for both sales channels. These measures provide the baseline for subsequent product, territory and time-based analysis.

Key metrics evaluated:

- Sales Amount
- Total Product Cost
- Gross Profit
- Gross Margin %
- Transaction/record volume

3.3 EDAs

3.3.1 Sales & Profitability by Channel



3-1: Sales Performance by Channel

- Compares total sales generated through the Internet and Reseller channels.
- The chart highlights the relative contribution of each channel to overall sales.

Key observation:

- Reseller sales are substantially higher than Internet sales in the analyzed dataset.
- The result indicates a stronger revenue contribution from the reseller channel.

3.3.2 Monthly Sales Trend



3-2: Monthly Sales Trend

- Shows the movement of combined Internet and Reseller sales across the available monthly periods.
- Highlights periods of higher and lower sales activity, including the highest monthly sales recorded in November 2013.

Monthly sales were aggregated using OrderDate across the cleaned Internet and Reseller sales datasets. The analysis covered 38 months from December 2010 to January 2014. Annual sales increased from 2011 through 2013, while November 2013 recorded the highest monthly sales. January 2014 was treated as a partial period and therefore was not interpreted as a full-year decline.

More EDA is available on GIT Jupyter notebook.

3.4 EDA Summary

- Exploratory analysis was performed on the cleaned sales datasets.
- Channel-level analysis showed the relative contribution of Internet and Reseller sales.
- Monthly analysis identified the overall sales pattern across the available period.
- Product-category analysis provided an initial view of sales concentration across product groups.
- The validated cleaned datasets were then exported from Python for database implementation.

4 MySQL Data Modeling & Database Preparation

4.1 Data Modeling

4.1.1 Objective

- A relational analytical model was designed to separate transactional sales data from descriptive business attributes.
- Two fact tables were retained for Internet and Reseller sales, supported by shared dimension tables.

4.1.2 Model structure

- FactInternetSales contains Internet sales transactions.
- FactResellerSales contains Reseller sales transactions.
- Customer, Product, Employee, Reseller and Sales Territory dimensions provide descriptive attributes.
- Geography supports geographic analysis for customers and resellers.
- Employee hierarchy is represented through ParentEmployeeKey.

The data model illustrates how the two sales fact tables connect to shared and channel-specific dimensions through primary–foreign key relationships.

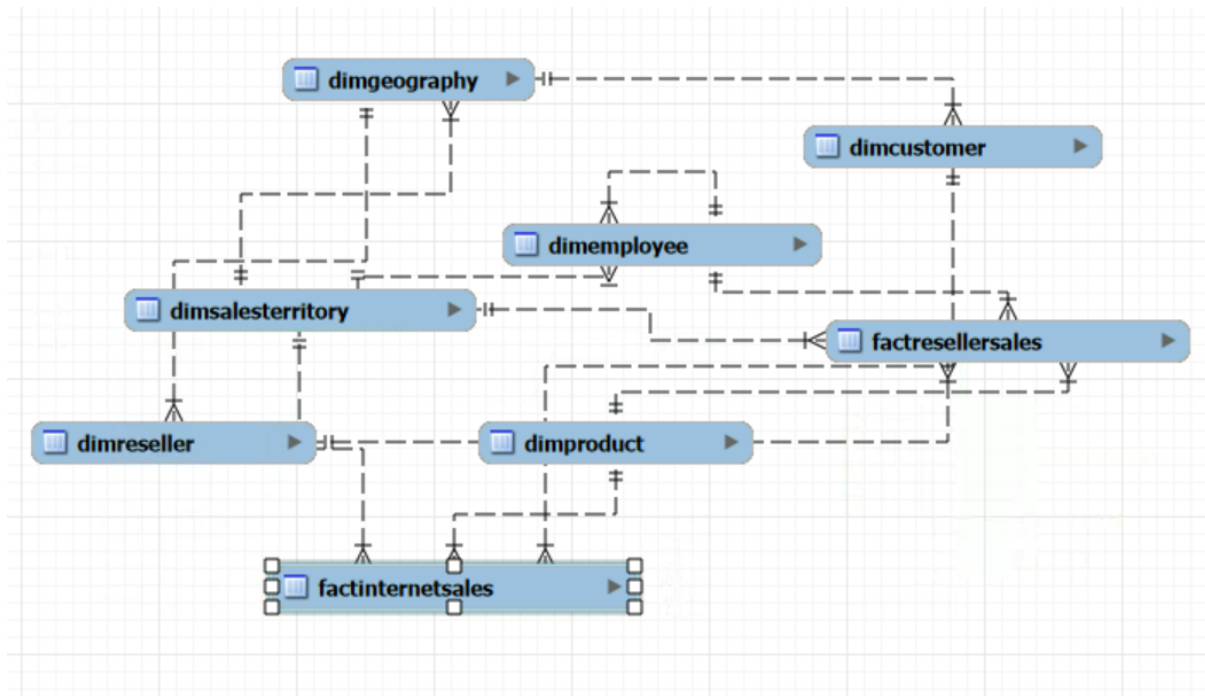
It provides the structural foundation for subsequent SQL analysis and Power BI reporting.

4.1.3 Data Model Design

A relational analytical model was designed using separate fact and dimension tables. Internet and reseller transactions are maintained as separate fact tables, while common business dimensions such as Product and Sales Territory support consistent analysis across sales channels. Customer and Reseller entities are linked to Geography, while Employee also maintains a self-referencing reporting hierarchy.

4.1.4 Relationship Validation

- All fact-table foreign keys were validated against their corresponding dimension keys.
- No invalid foreign-key values were identified.
- Both sales fact tables contained zero duplicate SalesOrderNumber + SalesOrderLineNumber combinations after cleaning.
- The employee hierarchy contained one top-level employee with a null ParentEmployeeKey; no invalid parent references were found.
- Dimension members without transactions were retained to preserve the complete business reference data.



4-1:SQL Data Model

- The model shows the relationship between sales fact tables and their supporting dimension tables through primary and foreign keys.
- The structure separates transactional measures from descriptive attributes, providing a foundation for SQL analysis and Power BI reporting.

4.2 MySQL Database Creation

4.2.1 Objective

Convert the cleaned datasets into a relational MySQL database with:

- 6-dimension tables
- 2 fact tables
- Primary keys
- Foreign keys
- Appropriate data types
- Referential integrity

	Tables_in_adventureworks_analytics
►	dimcustomer
	dimemployee
	dimgeography
	dimproduct
	dimreseller
	dimsalesterritory
	factinternetsales
	factresellersales

4-2:Table Creation

- The MySQL database contains six-dimension tables and two sales fact tables representing the cleaned analytical data model.
- Primary and foreign-key constraints establish the relationships required for consistent transactional and dimensional analysis.

	TotalRows	UniqueKeys	NullKeys
►	11	11	0

4-3:DimSalesTerritory data import validation

- The validation confirms that all 11 cleaned sales-territory records were successfully loaded into MySQL.

4.2.2 Database Loding

	TableName	RowCount
	DimCustomer	18484
	DimEmployee	21
	DimGeography	655
	DimProduct	397
	DimReseller	701
	DimSalesTerritory	11
	FactInternetSales	60398
▶	FactResellerSales	60855

4-4:MySQL database table and row-count validation

- Confirms that all eight analytical tables were successfully loaded into the MySQL database.
- Row counts match the cleaned datasets prepared during the Python data-validation stage.

5 Business SQL Analysis

5.1 Annual Sales Performance

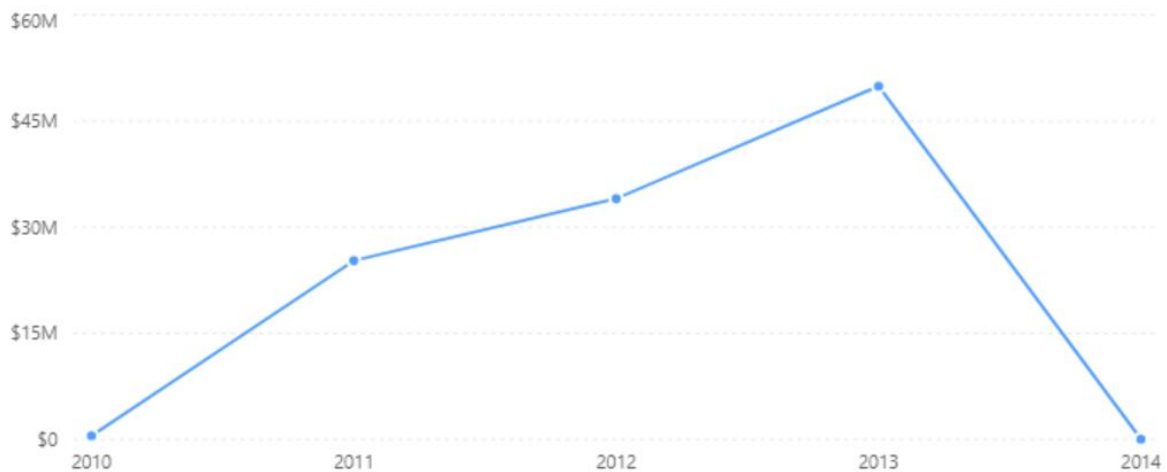
	SalesYear	InternetSales	ResellerSales	TotalSales
▶	2010	43421.04	489328.58	532749.62
	2011	7075525.93	18192802.71	25268328.64
	2012	5842485.20	28193631.53	34036116.73
	2013	16351550.34	33574834.16	49926384.50
	2014	45694.72	0.00	45694.72

5-1:Annual sales performance

- Total sales increased substantially from \$0.53M in 2010 to \$49.93M in 2013, with 2013 recording the highest annual sales in the available data.
- Internet sales became much stronger in 2013, while reseller sales remained the larger contributor overall through the major sales years.

Annual sales performance

Total sales across Internet and Reseller channels from 2010 to 2014.



5-2:Annual sales performance by sales channel

- Compares annual sales generated through Internet and Reseller channels.
- Shows the strong increase in total sales through 2013 and the partial-year nature of the 2014 data.

5.2 Sales by Product Category

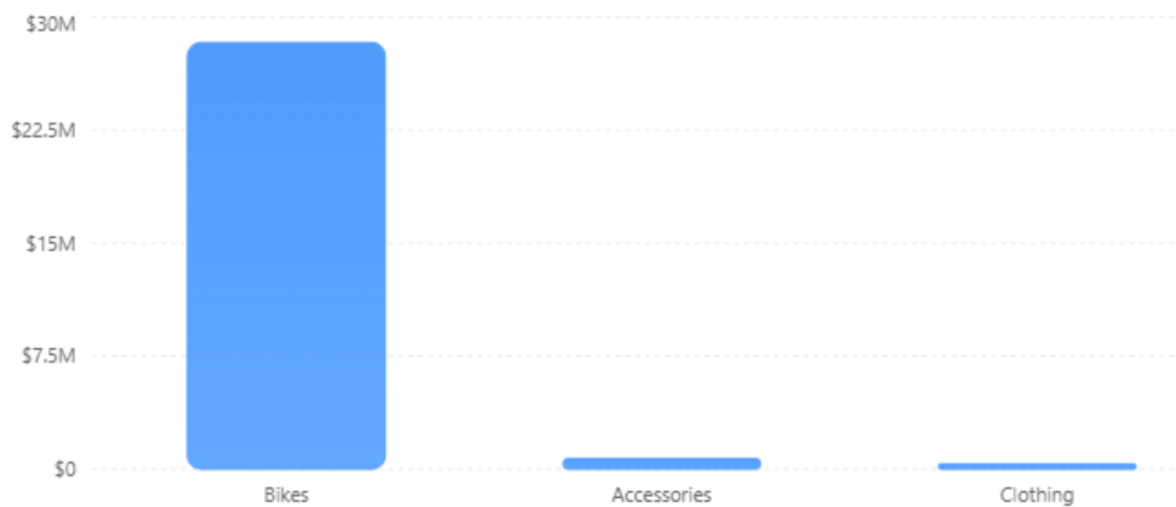
5-1: Internet sales performance by product category

Category	Total Sales	Products Sold	Sales Lines
Bikes	\$28,318,144.65	116	15,205
Accessories	\$700,759.96	22	36,092
Clothing	\$339,772.61	20	9,101

- Bikes generated by far the highest Internet sales, with approximately \$28.32M, compared with less than \$1M for either Accessories or Clothing.
- Accessories generated the highest number of sales lines (36,092) despite having much lower total sales, indicating a high volume of lower-value transactions.

Internet sales by product category

Total Internet sales generated by each product category.



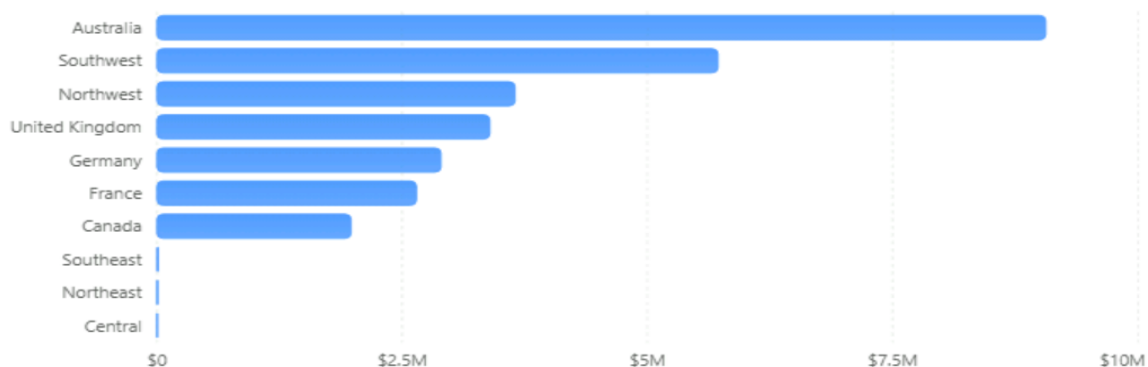
5-3: Internet sales performance by product category

- Compares sales value, number of products sold, and sales-line volume across product categories.
- Bikes dominate sales value, while Accessories have the highest transaction-line volume.

5.3 Sales by Territory

Internet sales by territory

Internet sales contribution across sales territories.



5-4: Internet sales performance by sales territory

- Australia is the leading Internet sales territory, generating approximately \$9.06M, followed by the U.S. Southwest at approximately \$5.72M.
- The remaining U.S. territories shown contribute very small sales values compared with the major territories, creating a substantial geographic difference in Internet sales performance.

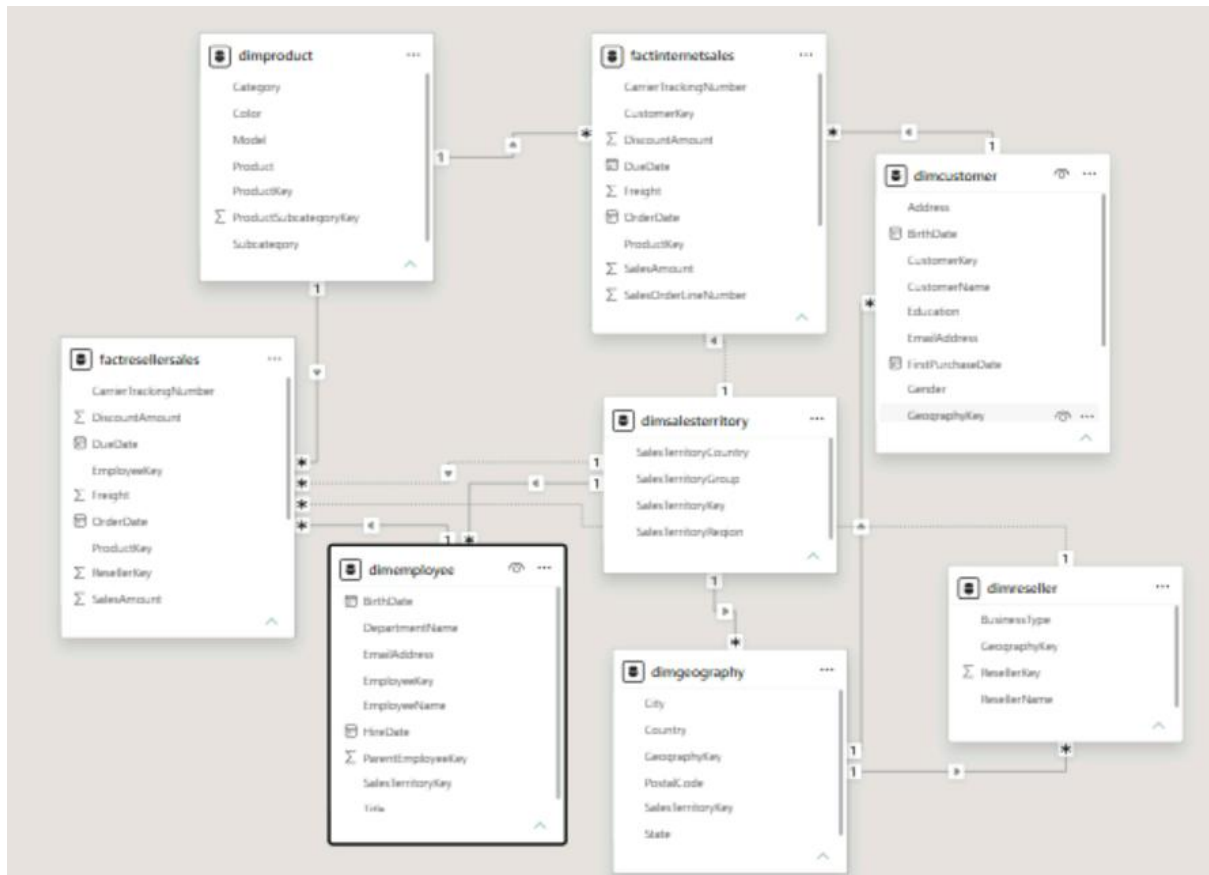
5.4 Top 10 Products

5-2: Top 10 products by Internet sales

RANK	PRODUCT	SUBCATEGORY	SALES
1	Road-150 Red, 48	Road Bikes	\$1,205,876.99
2	Road-150 Red, 62	Road Bikes	\$1,202,298.72
3	Road-150 Red, 52	Road Bikes	\$1,080,637.54
4	Road-150 Red, 56	Road Bikes	\$1,055,589.65
5	Road-150 Red, 44	Road Bikes	\$1,005,493.87
6	Mountain-200 Black, 42	Mountain Bikes	\$979,960.73
7	Mountain-200 Silver, 38	Mountain Bikes	\$979,035.78
8	Mountain-200 Black, 46	Mountain Bikes	\$961,600.81
9	Mountain-200 Black, 38	Mountain Bikes	\$954,715.84
10	Mountain-200 Silver, 46	Mountain Bikes	\$930,315.99

- The Road-150 Red variants occupy the top five positions, with the 48-inch variant generating the highest Internet sales at \$1.21M.
- All 10 top-selling products belong to the Bikes category, reinforcing the strong contribution of bikes to overall Internet sales.

6 Power BI Data Model & Dashboard



6-1:Power BI analytical data model

Shows the eight tables imported from the validated MySQL analytical database.

6.1 Power BI Sales Performance Dashboard



6-2:Power BI Sales Performance Dashboard

- The dashboard combines sales, profitability, order, product, territory, and sales-channel analysis into a single interactive reporting view.
- Date and territory filters allow users to interactively examine sales performance across different periods and business regions.

7 Findings & Business Insights

7.1 Key Business Findings

7.1.1 Reseller channel is the dominant sales contributor

Total sales across the complete dataset are approximately \$109.81M, consisting of:

Internet Sales: \$29.36M

Reseller Sales: \$80.45M

Reseller sales therefore contribute the larger share of overall revenue.

Business insight: The reseller channel is currently the primary contributor to sales and should remain a major focus for revenue retention and growth.

7.1.2 2013 was the strongest full year

Annual combined sales increased to approximately \$49.93M in 2013, making it the highest-sales year in the dataset.

YEAR	TOTAL SALES
2010	\$0.53M
2011	\$25.27M
2012	\$34.04M
2013	\$49.93M
2014*	\$0.05M

*2014 contains only a partial period.

Business insight: The strong 2013 performance indicates significant growth compared with the preceding years, particularly through the reseller channel and increasing Internet Sales.

7.1.3 Bikes dominate Internet Sales

Internet Sales by product category show a very strong concentration:

- Bikes: \$28.32M
- Accessories: \$0.70M
- Clothing: \$0.34M

Bikes account for the overwhelming majority of Internet Sales value.

Business insight: Bike products are the primary revenue-driving category in the online channel. Product planning, inventory availability, and promotional efforts should therefore give particular attention to high-performing bike products.

7.1.4 Australia is the strongest Internet Sales territory

Australia generated approximately \$9.06M in Internet Sales, followed by:

- Southwest US: \$5.72M
- Northwest US: \$3.65M
- UK: \$3.39M
- Germany: \$2.89M
- France: \$2.64M
- Canada: \$1.98M

Business insight: Australia is the strongest Internet market in the analyzed data. The substantial difference between leading and lower-performing territories suggests opportunities for targeted regional strategies.

7.1.5 Road-150 products are particularly strong performers

The Top 10 Internet products are dominated by Bike products, with several Road-150 variants occupying the highest positions.

The strongest individual product was:

- Road-150 Red, 48 — approximately \$1.21M

The top five positions are Road-150 Red variants.

Business insight: High-performing Road-150 variants appear to be important contributors to Internet Sales and could warrant closer attention for inventory planning and product promotion.

7.1.6 Overall profitability is positive

The Power BI model produces:

- Total Sales: \$109.81M
- Total Product Cost: approximately \$97.26M
- Gross Profit: approximately \$12.55M
- Gross Profit Margin: 11.43%

Business insight: The business generates positive gross profit, although the approximately 11.4% gross margin indicates that cost management and product/channel profitability remain important areas for management attention.

7.2 Business Recommendations

Based on the validated findings, the following recommendations can be included in the report:

- **Strengthen the reseller channel** while continuing to develop Internet Sales because reseller sales currently represent the largest revenue contribution.
- **Prioritize high-performing bike products**, particularly strong Road-150 variants, for inventory planning and promotional activity.
- **Investigate regional differences** in Internet Sales and develop targeted strategies for lower-performing territories rather than applying a uniform approach across all markets.
- **Continue developing the Internet channel**, given the significant improvement observed in 2013.
- **Monitor gross margin alongside sales growth**, ensuring that revenue growth is not achieved at the expense of profitability.